Customer Segmentation and Revenue Forecasting in E-Commerce: A Case Study

Abstract
This paper investigates whether customer segmentation improves the accuracy of revenue forecasting for an online retail business. We combine clustering techniques with time-series forecasting to test this hypothesis.

1. Introduction and Objective
The main objective of this research was to analyze customer purchasing patterns and improve revenue forecasting for an e-commerce platform. Prior work has treated all customers as a single homogeneous group when forecasting revenue, which we hypothesize loses important signal.

2. Dataset
We used the Online Retail dataset, a publicly available transactional dataset from a UK-based online retailer. The dataset contained approximately 500,000 transaction records spanning December 2010 to December 2011, including invoice number, product code, quantity, invoice date, unit price, customer ID, and country.

3. Methodology
Our methodology used a combination of clustering and time-series forecasting. We first applied K-means clustering to group customers into segments based on recency, frequency, and monetary value (RFM analysis). We then trained a separate time-series forecasting model (SARIMA) for each customer segment and compared the aggregated forecast to a single non-segmented baseline model.

4. Results
The segmented forecasting approach reduced mean absolute percentage error (MAPE) by 14% compared to the non-segmented baseline. High-value customer segments showed the most predictable purchasing cycles, while low-frequency segments contributed most of the forecasting error in both models.

5. Conclusion
Customer segmentation improved forecast accuracy. We conclude that treating customer segments separately, rather than forecasting revenue in aggregate, produces measurably better forecasts and is a low-cost improvement for e-commerce businesses already collecting transaction data.

6. Limitations
This study used data from a single retailer in a single country, so results may not generalize to other markets. Future work should test this approach on a more diverse dataset.